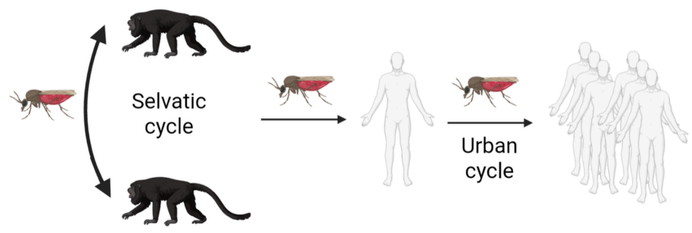
De novo computational design of epitope-specific nanobodies against the *Orthobunyavirus oropoucheense* nucleocapsid for point-of-care diagnostics

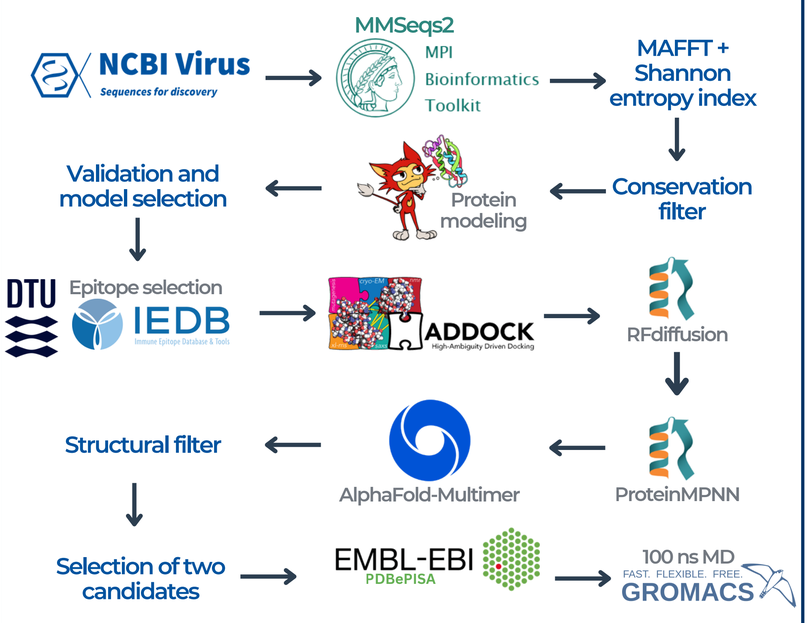
Alcántara-Ríos, C.[1] ; Hernandez, A.[1] ; Sandoval, G.[1]
[1] Faculty of Biological Sciences, Universidad Nacional Mayor de San Marcos, Peru

Background

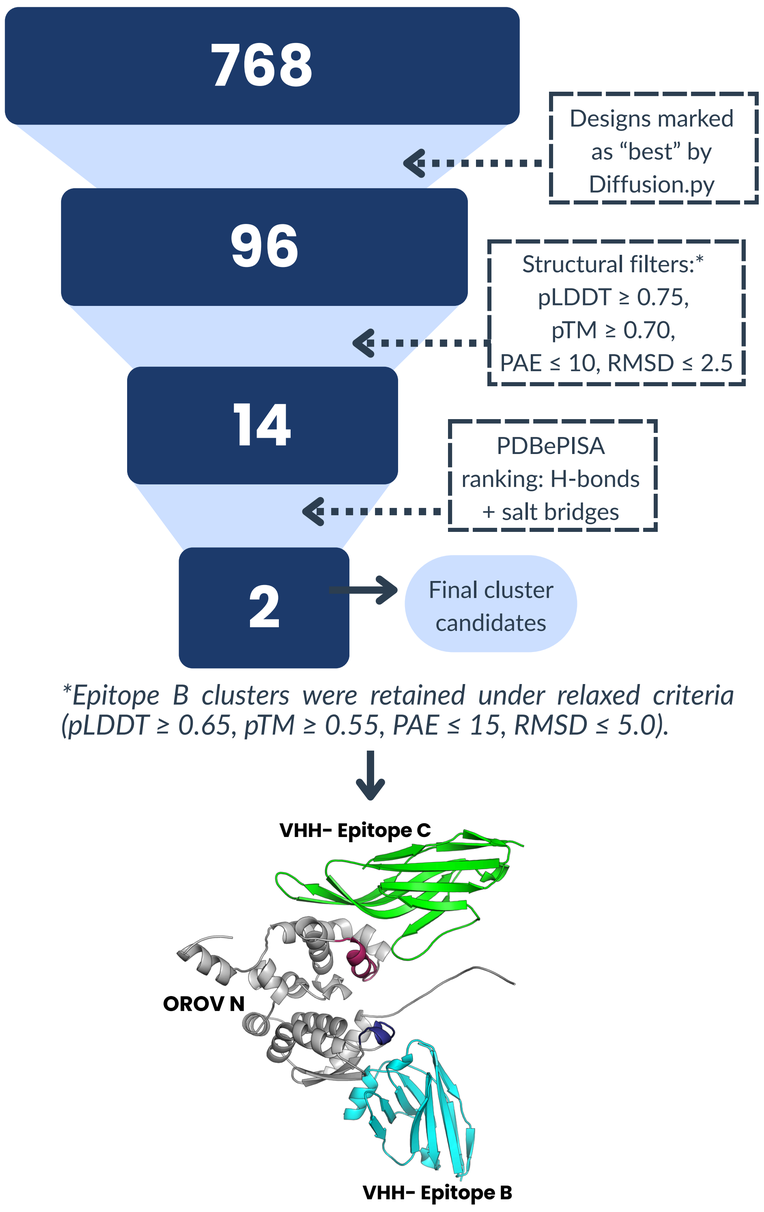
- The 2023-2025 Oropouche outbreak, with over 23,000 cases across the American continent, exposed critical public health gaps.
- Although RT-qPCR remains the diagnostic gold standard, it requires specialized infrastructure inaccessible in endemic rural areas, underscoring the urgent need for low-cost point-of-care (POC) alternatives such as lateral flow assays.
- Traditional nanobody discovery relies on animal immunization, which is slow and unable to target predefined epitopes, and no epitope-directed POC reagents have been reported for the *Orthobunyavirus oropoucheense* species complex.



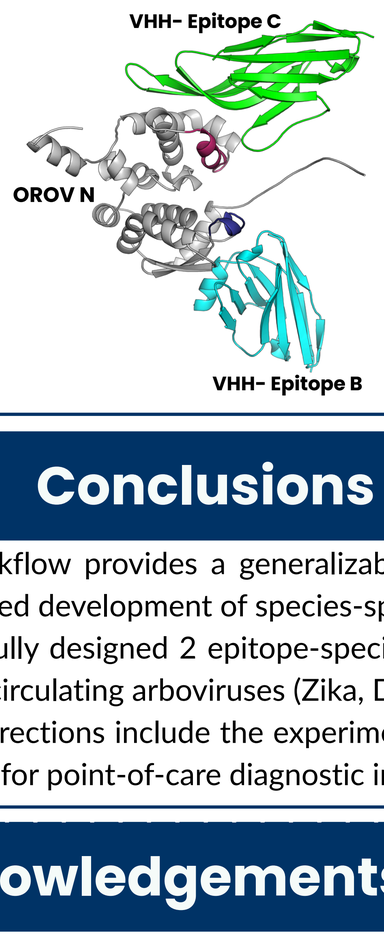
Methodology



Selection funnel



**Epitope B clusters were retained under relaxed criteria (pLDDT ≥ 0.65, pTM ≥ 0.55, PAE ≤ 15, RMSD ≤ 5.0).*

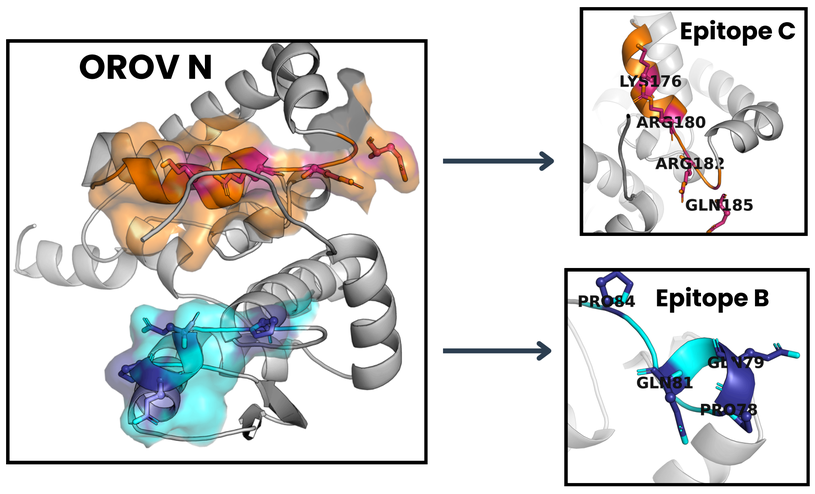


Results

Two Pan-Specific Epitopes Identified

Epitope B
- Shannon entropy: 0.01290 (highly conserved)
- Moderate RNA-binding interference
- SASA: 23.99-117.93 Å² per residue

Epitope C
- Shannon entropy: 0.01667 (highly conserved)
- Low RNA-binding interference (exposed epitope)
- SASA: 0.80-170.70 Å² per residue



Exclusivity analysis

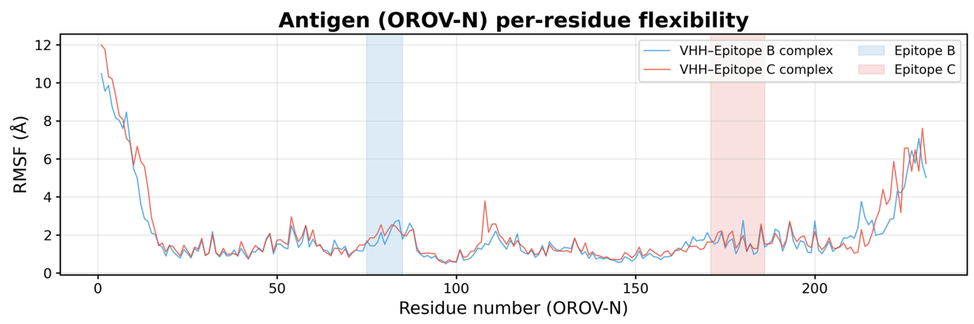
Table 1. Local Sequence Identity Against Simbu Serogroup Homologs

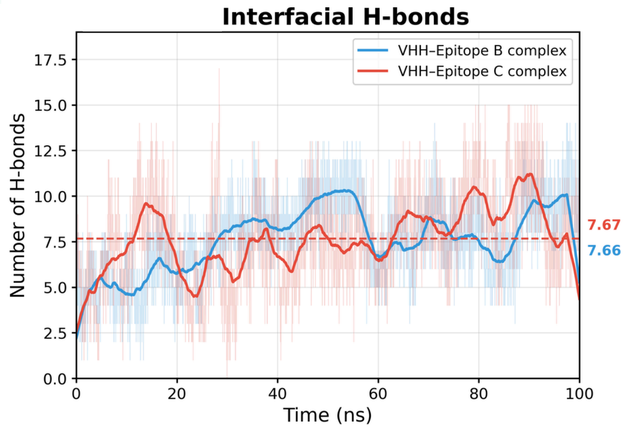
Evaluated Virus	Epitope B Identity	Epitope C Identity	Cross-Reactivity Risk
Schmallenberg virus	81.81%	75.00%	Medium–Low (Null due to geographic restriction)
Akabane virus	72.72%	87.50%	Medium–Low (Null due to geographic restriction)
Jatobal virus	90.90%	100.00%	**Very High**
Manzanilla virus	100.00%	81.25%	**High**
Bunyamwera virus	45.45%	31.25%	**Low–Null**

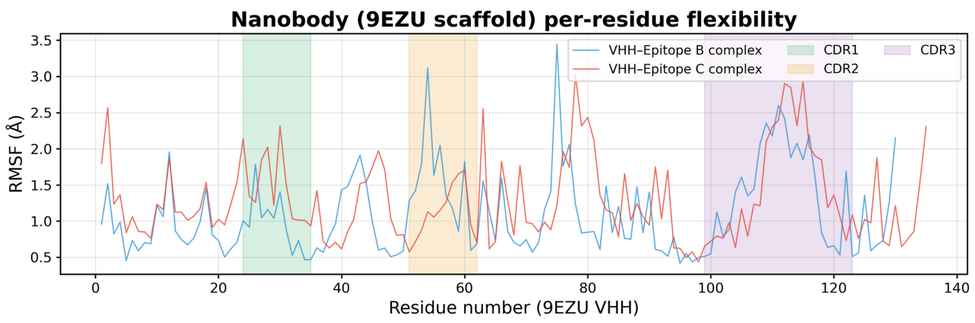
Table 2. *In Silico* Exclusivity Analysis (BLASTp) Against Co-Circulating Arboviruses

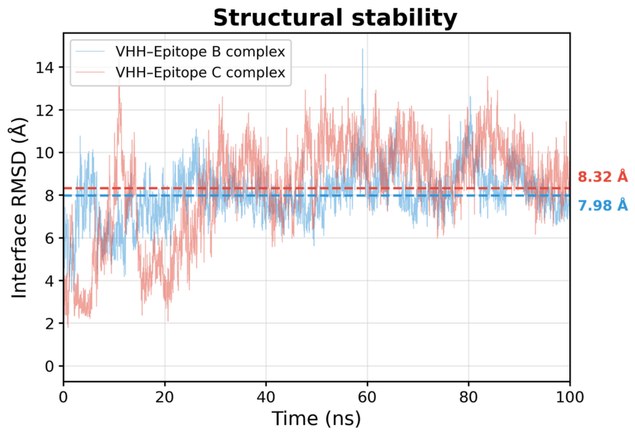
Evaluated Virus	Viral Family	E-value (Epitope B)	E-value (Epitope C)	Cross-Reactivity Risk
Zika virus	*Flaviviridae*	>1.0	>1.0	**Null**
Dengue virus	*Flaviviridae*	>1.0	>1.0	**Null**
Chikungunya virus	*Togaviridae*	>1.0	>1.0	**Null**
Mayaro virus	*Togaviridae*	>1.0	>1.0	**Null**
Hantavirus	*Hantaviridae*	>1.0	>1.0	**Null**

100 ns molecular dynamics simulations









Conclusions

- This workflow provides a generalizable, epitope-directed framework for rapid, low-cost nanobody design against emerging arboviruses, supporting the accelerated development of species-specific POC lateral flow assays for resource-limited settings in Latin America.
- Successfully designed 2 epitope-specific nanobodies *de novo* showing high sequence conservation across the Oropouche complex and no cross-reactivity with co-circulating arboviruses (Zika, Dengue, Chikungunya, and Mayaro).
- Future directions include the experimental validation (in vitro) of these designed nanobodies and the expansion of the screening library to optimize binding affinities for point-of-care diagnostic integration

References

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